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A Data-Driven Framework for Assessing Sustainability-Oriented Research Models in Higher Education Institutions

Marco Ruben Burbano Pulles *  and Jhonatan Bladimir Cuadrado Merlo * 

Facultad de Industrias Agropecuarias y Ciencias Ambientales UPEC, Universidad Politécnica Estatal del Carchi, Calle Antisana y Av. Universitaria, Tulcán 040101, Ecuador

* Correspondence: marco.burbano@upec.edu.ec (M.R.B.P.); jhonatan.cuadrado@upec.edu.ec (J.B.C.M.)

Abstract

The growing demand for sustainable development has exposed significant limitations in how Higher Education Institutions (HEIs) evaluate sustainability-oriented research, particularly due to fragmented indicators, descriptive approaches, and the absence of robust data-driven assessment frameworks. This study proposes a comprehensive framework for evaluating the sustainability orientation of university research models, integrating validated measurement instruments with advanced analytical and predictive techniques to support evidence-based decision-making in higher education governance. The framework is based on a multidimensional instrument comprising 26 indicators across environmental, social, economic, and institutional dimensions, developed through expert judgment using the Delphi method and statistically validated through Confirmatory Factor Analysis (CFA). The instrument was applied to 260 researchers from four public HEIs located in the Colombia–Ecuador border region, and perceived performance was contrasted with actual institutional indicators, revealing significant nonlinear discrepancies. To address this complexity, an artificial neural network model was developed to estimate real sustainability performance based on survey data, achieving a predictive accuracy of 90.92%. Beyond institutional diagnosis, the proposed framework functions as a decision-support tool that enables HEIs to identify critical gaps, prioritize interventions, and guide continuous improvement strategies in research management. Due to its methodological rigor, scalability, and transferability, the framework can be adapted to diverse higher education contexts, contributing to the advancement of sustainability assessment methods and governance practices in universities.

Keywords: sustainability assessment; higher education; research sustainability; data-driven evaluation; confirmatory factor analysis; artificial neural networks; sustainability governance



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1. Introduction

Higher Education Institutions (HEIs) play a central role in advancing sustainable development through knowledge generation, innovation, and the transfer of research out-comes to society. Within the framework of the Sustainable Development Goals (SDGs), universities are increasingly expected to align their research systems with environmental protection, social equity, economic resilience, and institutional responsibility [1–6]. Consequently, sustainability has become a strategic dimension of university governance, requiring systematic evaluation mechanisms that allow institutions to monitor, assess, and improve the sustainability orientation of their research models [7–10].

Despite growing recognition of the importance of sustainability in higher education, significant challenges persist in evaluating university research systems. Existing approaches often rely on isolated indicators, descriptive reports, or administrative metrics that fail to capture the multidimensional and dynamic nature of sustainability [9–11]. Moreover, many proposed frameworks remain conceptual or lack rigorous statistical validation, limiting their usefulness for evidence-based decision-making [11–13]. These limitations are particularly evident in public HEIs operating in complex territorial contexts, where sustainability challenges are intensified by social, economic, and environmental constraints [6,14].

Recent research highlights the need for integrated, data-driven frameworks capable of evaluating sustainability performance across multiple dimensions within higher education systems [8–11,14]. Such approaches should combine robust measurement instruments with advanced analytical methods to reduce subjectivity, capture nonlinear relationships, and support continuous improvement [10,12,13,15]. Multivariate statistical techniques, particularly Confirmatory Factor Analysis (CFA), have proven effective in validating sustainability-related constructs and ensuring measurement reliability in higher education research [12,13,15,16].

In parallel, the increasing availability of institutional data has encouraged the application of artificial intelligence and machine learning techniques in sustainability assessment and university governance [17–22]. Artificial neural networks, in particular, provide strong capabilities for modeling complex nonlinear relationships between perceptions, behaviors, and institutional performance, offering predictive power beyond traditional linear approaches [19,20,23]. When integrated with validated measurement instruments, these techniques enable the development of decision-support systems that transform empirical data into actionable insights for sustainability-oriented management [16,21,22].

From an assessment perspective, university research models represent a critical unit of analysis within HEIs. Research policies, funding structures, academic practices, and societal engagement directly influence the contribution of universities to sustainable development [3,4]. However, validated tools specifically designed to evaluate sustainability-oriented research models across environmental, social, economic, and institutional dimensions remain limited, particularly in the context of Latin American public universities [9,11,24–26].

To address these gaps, this study proposes a data-driven framework for evaluating sustainability-oriented research models in Higher Education Institutions. The framework integrates a multidimensional instrument developed through expert judgment and validated using Confirmatory Factor Analysis with an artificial neural network model designed to predict actual sustainability performance based on researchers' perceptions. The empirical application focuses on four public HEIs located in the Colombia–Ecuador border region, a territory characterized by significant sustainability challenges and strong demand for socially relevant research [2,6].

The objectives of this study are threefold: (i) to construct and statistically validate an instrument for assessing the sustainability orientation of university research models; (ii) to analyze discrepancies between perceived and actual institutional performance using multivariate techniques; and (iii) to develop a predictive model that supports evidence-based decision-making in university research governance. By combining explanatory and predictive approaches, the proposed framework extends beyond descriptive evaluation and provides a practical decision-support tool for sustainability-oriented management.

The main contribution of this research lies in the development and validation of a data-driven framework that integrates a statistically validated multidimensional instrument with advanced analytical techniques, including Confirmatory Factor Analysis and artificial neural networks. This approach enables the identification of discrepancies be-

tween perceived and actual sustainability performance while providing a scalable and transferable tool adaptable to diverse institutional and territorial contexts, contributing to the advancement of sustainability assessment methodologies in higher education [9,10].

1.1. Related Works

Research carried out by Higher Education Institutions (HEIs) plays a leading and indispensable role in the sustainable development of societies [1]. As noted by [2,3], university research is key to generating the knowledge, innovations, and solutions required to promote environmentally sustainable, socially inclusive, and economically vibrant progress. Therefore, it is essential for HEIs to have relevant research models that effectively guide their research capacities to strategically address the needs and priorities of their context [4].

In the current context, relevance in university research models implies adopting a sustainability-oriented approach in a transversal and holistic manner. As stated by [5,6], sustainability must be embedded in all areas of HEIs, including their research systems. This requires research models to incorporate objectives, policies, procedures, and practices that explicitly promote the generation of knowledge for sustainable development in its multiple dimensions [7].

The core components of university research models must be tangibly linked to concrete sustainability indicators, considering their environmental, social, economic, and institutional dimensions in an integrated manner [8]. Aspects such as funding, human resource allocation, infrastructure, scientific production processes, and dissemination and social engagement activities, among others, must be intentionally aligned with clearly specified sustainability goals and indicators [9]. Only in this way can university research optimally contribute to addressing pressing challenges such as climate change, social inequalities, poverty, and biodiversity loss.

For this reason, it is valuable and strategic to have comprehensive tools that allow for the evaluation of the actual level of compliance of university research models with concrete sustainability indicators across their environmental, social, economic, and institutional dimensions. Such instruments make it possible to conduct a detailed and evidence-based diagnosis of the strengths and weaknesses of existing models [10], and subsequently to propose well-founded and specific improvements that optimally adjust these models, so that university research is oriented toward sustainability in the most effective way [11]. In addition, these tools enable the monitoring and comparison of progress among different HEIs over time [14]. Undoubtedly, the availability of robust instruments to evaluate and strengthen the incorporation of sustainability into university research systems constitutes a relevant contribution, which could be replicated and adapted to optimize research models in diverse contexts.

1.2. Literature Review and Theoretical Framework

The evaluation of sustainability in Higher Education Institutions (HEIs) has gained increasing attention over the past decade, particularly as universities assume a more active role in addressing global sustainability challenges. Within the framework of the Sustainable Development Goals (SDGs), universities are expected not only to generate knowledge but also to incorporate sustainability principles into governance, teaching, research, and societal engagement [1,2]. As a result, sustainability assessment has emerged as a critical tool for monitoring institutional performance and guiding strategic decision-making in higher education [5,9].

Several approaches have been proposed to evaluate sustainability in HEIs, ranging from indicator-based frameworks to composite indices and reporting systems [9,10]. How-

ever, many of these approaches rely on fragmented indicators or descriptive methodologies that fail to capture the multidimensional and systemic nature of sustainability [7,11]. In particular, the evaluation of sustainability-oriented research remains underdeveloped, despite its central role in knowledge production and innovation for sustainable development [3,4]. This gap highlights the need for integrated and empirically validated assessment frameworks capable of capturing environmental, social, economic, and institutional dimensions simultaneously.

From a methodological perspective, the literature emphasizes the importance of using statistically validated measurement models to ensure reliability and construct validity in sustainability assessment [12,13]. Confirmatory Factor Analysis (CFA) has been widely used to validate multidimensional constructs and to establish robust relationships between observed indicators and latent sustainability dimensions in higher education research [10,12]. These approaches contribute to improving the objectivity and consistency of sustainability measurement while enabling comparative institutional analysis.

More recently, advances in data analytics and artificial intelligence have opened new opportunities for modeling complex sustainability dynamics in higher education systems. Machine learning techniques, particularly artificial neural networks, have demonstrated strong capabilities in capturing nonlinear relationships between institutional indicators, stakeholder perceptions, and performance outcomes [17–20]. Unlike traditional linear models, neural networks can incorporate multidimensional and interdependent variables, providing predictive insights that support evidence-based governance and decision-making in sustainability-oriented management.

From a theoretical standpoint, this study adopts a multidimensional view of sustainability in higher education, conceptualizing university research models as systems influenced by structural, organizational, and societal factors. Research policies, funding mechanisms, academic practices, and institutional engagement collectively shape the contribution of universities to sustainable development [3,4]. Within this perspective, sustainability assessment is understood not merely as a descriptive exercise but as an analytical process that integrates measurement validation, structural analysis, and predictive modeling to support continuous institutional improvement.

Based on this theoretical and methodological foundation, the present study develops a data-driven framework that integrates validated measurement instruments with advanced analytical techniques, including Confirmatory Factor Analysis and artificial neural networks, to evaluate sustainability-oriented research models in Higher Education Institutions.

2. Materials and Methods

The development of the stages of this research was based on the Delphi methodology, applied under a classical two-round structure following established methodological guidelines [27], aimed at identifying, refining, and reaching consensus on sustainability indicators for research models in HEI. In the first round, qualitative input was obtained from a panel of experts through an initial questionnaire composed of four open-ended questions designed to identify, through expert judgment, the indicators relevant to each sustainability dimension. These questions were formulated with a diagnostic perspective to capture the current state of research practices, structures, and outcomes related to sustainability in the participating institutions. The responses were systematically analyzed and synthesized to identify patterns and areas of convergence. In the second round, the refined set of indicators was evaluated by a broader group of researchers from different HEIs, allowing consensus building and strengthening the content validity of the instrument. This iterative process ensured methodological rigor and alignment with standard Delphi practices.

The recruitment phase initially involved experts belonging to the research group Sociedad Sustentable at the Universidad Politécnica Estatal del Carchi, who participated in the first Delphi round. Subsequently, in the second round, researchers from different Higher Education Institutions, both nationally and internationally, were incorporated in order to consolidate consensus and strengthen the validity of the instrument, following standard Delphi methodological guidelines [27]. Based on this process, a final matrix integrating the sustainability dimensions and their respective indicators was constructed, which served as the basis for developing a structured instrument composed of 26 ordinal Likert-scale questions.

In the instrument design phase, the initial questionnaire presented in Table 1 consisted of four open-ended questions aimed at identifying, through expert judgment, the indicators relevant to each sustainability dimension (environmental, social, economic, and institutional) within university research models. These questions formed part of the first Delphi round and were designed with a diagnostic orientation to describe the current situation of sustainability-related practices and structures in the research systems of the analyzed HEIs. The responses obtained were analyzed and synthesized to identify patterns, convergences, and common elements among experts, enabling the construction of a preliminary matrix of sustainability indicators. This matrix was subsequently refined through the second Delphi round, in which indicators were consolidated through expert consensus, forming the basis for the structured instrument presented in Table 2.

The instrument was applied to a total sample of 260 researchers from four public HEIs located in the Colombia–Ecuador border region. The empirical phase of the study was conducted over a defined period. Data collection was carried out using an online survey platform configured to ensure anonymity, controlled access, and standardized response recording. Participants were given a fixed response window to complete the questionnaire, during which reminder notifications were sent to maximize participation. A total of 260 valid responses were obtained from researchers across the four participating universities. Responses were reviewed for completeness and consistency prior to statistical analysis.

In addition to the Delphi consensus process, the responses obtained from the structured instrument presented in Table 2 were statistically validated to ensure construct validity and reliability. Confirmatory Factor Analysis (CFA) was employed to verify the factorial structure of the instrument and to evaluate the extent to which each observed variable adequately represented its corresponding latent dimension. Model fit was assessed using widely accepted goodness-of-fit indices, including the Comparative Fit Index (CFI), Tucker–Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA), and Standardized Root Mean Square Residual (SRMR), following established methodological standards [13,15,28–31]. The results confirmed satisfactory internal consistency and supported the validity of the instrument for representing the environmental, social, economic, and institutional dimensions measured through the responses summarized in Table 2.

Table 1. Questions formulated to identify indicators relevant to each dimension of sustainability.

Description	
Code	Question
p1	What should be the indicators that should form part of the social dimension for a research model of an HEI located in the Colombia–Ecuador border area?
p2	What percentage of the Research Department’s budget for the development of sustainability-oriented research activities comes from external funding?
p3	What percentage of the Research Department’s budget is used to grant incentives for research activities focused on sustainability?
p4	What is the number of sustainability-related research activities carried out ad honorem?

Table 2. Questions included in the instrument administered to researchers from the participating Higher Education Institutions (HEIs).

Economic Dimension	
Code	Question
p4	There is a considerable percentage of the Research Department's budget for the development of sustainability-oriented research activities coming from external funding.
p5	The percentage of the Research Department's budget used to grant incentives for research activities focused on sustainability is adequate.
p6	The IES develops a considerable number of ad honorem sustainability-related research activities per year.
p7	There is an adequate percentage of research projects involved in the optimization of economic resources through sustainable alternatives.
p8	There is an adequate percentage of research projects based on local and regional development with a sustainability focus.
Social dimension	
p9	There is an adequate number of departments and/or centers that carry out research activities providing services to society.
p10	There is a considerable number of sustainability reports and/or documents derived from research activities that are published by the IES and linked to society per year.
p11	There is equity in gender participation in research activities.
p12	There is an adequate percentage of research projects related to renewable energy, ecological economics, urban planning, or other aspects of an environmental nature.
Environmental dimension	
p13	There is an acceptable number of departments, units, or environmental management centers that conduct research.
p14	The IES organizes a significant number of scientific events oriented toward the environmental component per year.
p15	There is a significant percentage of research groups whose practice is focused on the environmental dimension.
p16	There is a significant percentage of research seedbeds that focus their work on the environmental dimension.
p17	The IES has a significant number of annual participations in external scientific events oriented toward the environmental component.
p18	The institutional mission and vision are linked to the principles of sustainability.
Institutional dimension	
p19	A considerable percentage of the IES's research faculty has training in sustainability.
p20	The percentage of students who graduate with research topics in sustainability in the degree or program in which you work is considerable.
p21	The institutional educational model includes curricular spaces oriented toward formative research under a sustainability approach.
p22	There is a considerable number of research publications per year focused on topics related to sustainability.
p23	There is an appropriate number of clubs focused on research and sustainability activities.
p24	There is an appropriate number of research groups focused on research and sustainability activities.
p25	There is an appropriate number of research seedbeds focused on research and sustainability activities.
p26	The percentage of postgraduate programs that have an approach exclusively based on sustainability is adequate.

The survey was conducted anonymously, ensuring the confidentiality of all participants through the configuration of the platform used for data collection, so that the responses provided by the experts remained anonymous throughout the entire process.

In the first instance, the initial questionnaire was sent to the selected experts for the corresponding round, establishing a defined period for its completion. Subsequently, the responses were collected and analyzed, carefully examining the preliminary opinions and assessments provided. Thereafter, the responses from the initial round were summarized, and the experts were provided with an anonymous compendium containing the responses and comments of the other participants.

Based on these results, the initial matrix was constructed, integrating the sustainability dimensions along with their respective indicators. During the application phase of this matrix, the responses were evaluated, consensually discussed, and adapted. The responses generated by the experts in the initial round were then sent to researchers from external HEIs in the form of an instrument consisting of 26 ordinal questions on a Likert scale, with the objective of incorporating their perceptions and contributions. Finally, the results were summarized and documented in a matrix that incorporated the recommendations and contributions of experts from external HEIs. The developed instrument is presented in Table 2.

After analyzing the indicators consolidated in the final matrix, their theoretical grounding within the research framework was strengthened, and two instruments derived from this matrix were structured. The first instrument was focused on obtaining specific and direct information from the representatives of the Research Departments of the evaluated HEIs, while the second instrument was aimed at exploring the degree of perceived compliance with each of the indicators by the surveyed researcher faculty members.

Data treatment. To facilitate interpretation, the mathematical formulation of the analytical procedures employed in this study is presented below. Each equation describes the statistical and computational steps used in the validation and modeling process. Variables, parameters, and matrices are defined immediately after each expression to ensure methodological clarity and reproducibility. The compilation of a database for any scientific study has the potential to contain missing and outlier data; therefore, it is noted that any statistical analysis must begin with the implementation of a meticulous data analysis protocol. Among the prevalent techniques for data handling in multivariate samples is the use of Mahalanobis distances. This technique facilitates the measurement of the number of standard deviations by which an observation is located with respect to the mean of a distribution. Since outliers do not behave in the same way as regular observations, this measure can be used for the detection of anomalous observations. From a geometric perspective, while Euclidean distance is the shortest distance between two points, it does not incorporate in its calculation the existing relationship among variables that are highly correlated. In contrast, Mahalanobis distance does take into account the correlation between variable [12,32]. This scale-invariant metric considers the measurement of the distance between a point $x \in \mathbb{R}^p$ drawn from a p -variate probability distribution $f_X(\cdot)$, p -varied and the mean $\mu = E(X)$ of the distribution. Under the assumption that the distribution $f_X(\cdot)$ has finite second-order moments, the covariance matrix can be determined as $\Sigma = E(X - \mu)$. Thus, the Mahalanobis distances are defined as:

$$D(X, \mu) = \sqrt{(X - \mu)^T \Sigma^{-1} (X - \mu)}. \quad (1)$$

where X represents the vector of observed values, μ denotes the vector of mean values, and Σ^{-1} corresponds to the inverse of the covariance matrix. The Mahalanobis distance was used to identify multivariate outliers and to ensure the robustness of the dataset prior to further statistical analysis.

Confirmatory factor analysis. Once the processed data were available, confirmatory factor analysis was applied to verify the validity and reliability of the instrument. Confirmatory factor analysis (CFA) is a technique used to test the variables that make up a construct, considering the dependence of each variable on the factor from which it originates. CFA stands out as an advantageous technique in the analysis of ordinal variables, through which it is possible to discern the extent to which the $p \times 1$ response vector constructed from observable random variables is capable of explaining one or more unobserved variables, known as factors η through this procedure, each proposed variable contributes to explaining the behavior of the unobserved latent variable that is evaluated and estimated. In this model, a vector of observed responses Y_i is considered to explain the unobserved latent variable ξ using the model.

$$Y = \Lambda \xi + \epsilon, \quad (2)$$

where Y represents the $p \times 1$ vector of observed random variables, ξ corresponds to unobserved variables, and Λ denotes a $p \times k$ matrix where k represents the number of unobserved latent variables, where k represents the number of unobserved latent variables. Since Y is composed of a set of unobserved latent variables ξ , the model includes the error term ϵ . This model can be solved using maximum likelihood (ML) estimation, generated through the iterative minimization of the function:

$$F_{ML} = \ln |\Lambda \Omega \Lambda' + I - \text{diag}(\Lambda \Omega \Lambda')| + \text{tr} \left(R \left(\Lambda \Omega \Lambda' + I - \text{diag}(\Lambda \Omega \Lambda')^{-1} \right) \right) - \ln(R) - p, \quad (3)$$

where $\Lambda \Omega \Lambda'$ represents the variance–covariance matrix derived from the proposed factor analysis model, and R corresponds to the observed variance–covariance matrix. More specifically, in Confirmatory Factor Analysis (CFA), the model parameters can be estimated by minimizing the distance between the variance–covariance matrix implied by the model and the observed matrix [13,15,28–31].

Principal Component Analysis (PCA) was employed as a dimensionality reduction technique, with the underlying objective of transforming a set of correlated variables into a set of uncorrelated variables. PCA has been commonly used as an exploratory data analysis technique, through which the existing relationships among a set of variables are examined, thus positioning it as a suitable technique for dimensionality reduction. This method, by seeking to transform an original number of possibly correlated variables into a smaller number of uncorrelated variables known as principal components, facilitates the interpretation of complex datasets by preserving, as much as possible, the total variance of the data. The first principal component is defined in such a way that it retains the maximum possible variance of the original data, and each subsequent component, in turn, captures the maximum remaining variance under the constraint of being orthogonal to the previous components [17]. In addition, as described in [18,19], PCA can be used to determine the number of hidden layers to be implemented in a neural network. For a dataset $x^{(1)}, x^{(2)}, \dots, x^{(m)}$ with n -dimensional observations, the aim is to project the dataset onto k -dimensional observations (where $k < n$). To do this, the process begins with the standardization of the data:

$$x_j^i = \frac{x_j^i - \bar{x}_j}{\sigma_j}, \quad (4)$$

Then, the covariance matrix is calculated as follows:

$$\Sigma = \frac{1}{m} \sum_i^m (x_i)(x_i)^T, \quad \Sigma \in \mathbb{R}^{n \times n} \quad (5)$$

Subsequently, the eigenvectors and eigenvalues of the covariance matrix are obtained through the equation:

$$U = \begin{bmatrix} | & | & | \\ u_1 & u_2 & \dots & u_n \\ | & | & | \end{bmatrix}, \quad u_i \in \mathbb{R}^n, \quad u^T \Sigma = \lambda \mu, \quad (6)$$

Consequently, the original data are projected into a k -dimensional subspace, in which the principal eigenvectors of the covariance matrix are selected. These new variables, or principal components, are able to represent the original data and their variance with reduced dimensionality. Each of these new vectors, or components, can be obtained through the expression:

$$x_i^{new} = \begin{bmatrix} u_1^T x^i \\ u_2^T x^i \\ \vdots \\ u_k^T x^i \end{bmatrix} \in \mathbb{R}^k \quad (7)$$

Neural networks, in their role as a classification technique, emerged as an ensemble method in which each artificial neuron sought to emulate the behavior of a biological neuron through the combination of a set of weights at its input. This mechanism allowed the neuron to activate and transmit a signal only if the combination of input signals reached a magnitude sufficiently high to surpass a predefined threshold. A wide range of activation functions that could govern the operation of each neuron was identified. However, in the present study, the function $ReLU \rightarrow o = \max(0, z)$ as selected for the design of the hidden layers, and the function $Softmax \rightarrow o = e^{z_i} / \sum_i e^{z_i}$ was selected for the output layer, which was required to exhibit binary behavior. Artificial neural networks were established as an ensemble technique in which it was possible to introduce as many input variables as necessary, through a neuron in the input layer that commonly did not include an activation function. Subsequently, as many connections as required were generated, assigning each connection a weight w_{ij} , which was a parameter assigned through the learning process. This allowed the various combinations of inputs to activate, or not, the neurons in the hidden and output layers, enabling each neuron, or combinations of them, to learn nonlinear behaviors from the data. The process of signal propagation in each layer of the neural network is obtained through the expression:

$$X_j = W_{ij} \cdot I, \quad (8)$$

$$O_j = \text{activation}(X_j), \quad (9)$$

In this context, X_j denotes the matrix of total input signals of the neurons in layer j of the neural network, while W_{ij} symbolizes the weight matrix of the connections between the current layer j and the previous layer i . Additionally, I is the matrix of input signals and O_j represents the matrix of output signals of each layer of the neural network. In order to enable the learning of a neural network, the error in each neuron of the final layer was calculated, $e_{out_k} = t_k - o_k$, by comparing the obtained value o_k with the expected value t_k for each observation t . These errors needed to be backpropagated through the neural network via the connections from which each output originated, thus allowing the weights to be updated. Errors can be backpropagated in the neural network using the expression:

$$\xi_i = W_{ij}^T \cdot \xi_j, \quad (10)$$

where ξ_i symbolizes the matrix of errors that were backpropagated to the preceding layer of the neural network, and ξ_j refers to the errors originating in the subsequent layer of the neural network. Once the errors were backpropagated through the neural network, these weights were used to allow the neural network to retain information from previous examples and incorporate new information derived from new observations. Gradient descent stood out as one of the most widely used processes for this purpose. One of the processes most commonly employed for this purpose is gradient descent, which is formulated as:

$$\frac{\partial \xi}{\partial W_{jk}} = \frac{\partial \sum_n (t_n - o_n)}{\partial W_{jk}} = \frac{\partial \xi}{\partial \mathcal{O}_k} \cdot \frac{\partial \mathcal{O}_k}{\partial W_{jk}} = -2(t_n - o_n) \cdot \frac{\partial \mathcal{O}_k}{\partial W_{jk}}, \quad (11)$$

$$\frac{\partial \xi}{\partial W_{jk}} = -2(t_n - o_n) \cdot \frac{\partial}{\partial W_{jk}} \text{activation} \left(\sum_j W_{jk} \cdot \mathcal{O}_j \right), \quad (12)$$

$$W_{jk}^{(r+1)} = W_{jk}^{(r)} - \alpha \frac{\partial \xi}{\partial W_{jk}}, \quad (13)$$

where $W_{jk}^{(r+1)}$ represents the new updated weight for a connection jk , which is updated based on its previous value $W_{jk}^{(r)}$ and the gradient $\partial \xi / \partial W_{jk}$, which provides a new portion of information moderated by the learning-rate hyperparameter α [20,33,34].

3. Results

The designed instrument was applied to a sample of 260 researchers from the four participating universities. In this way, each observation was composed of 23 ordinal variables that were structured as factors, as presented in Table 2, with the objective of enabling the evaluation of the four dimensions of sustainability applied to the educational models of each university. The descriptive statistics for each of the ordinal variables that comprised the survey are presented in Table 3.

Next, it was verified whether the collected database contained outlier observations; for this purpose, Mahalanobis distances were employed. This analysis was carried out using the mahalanobis function, and a cutoff score of 54.0519 was established based on the χ^2 distribution, considering 99.9% of the distribution as acceptable distances and excluding 0.1% of the observations as outliers. Through this procedure, 18 outlier observations were identified and removed from the database; therefore, the final database consisted of 242 multivariate observations.

With this new treated database, its parametric assumptions were verified, since the selected instrument validation technique was Confirmatory Factor Analysis (CFA), which requires compliance with the assumptions of additivity, normality, linearity, homogeneity, and homoscedasticity. In this way, the first assumption verified was additivity, for which the multivariate correlation matrix was used for each possible pair of ordinal variables. The correlation matrix for the sample used is presented in Figure 1.

As can be observed in Figure 1, when obtaining the bivariate correlation for each pair of questions, all of them reached an adequate correlation value and no pair of questions registered correlation values close to one; therefore, the assumption of additivity was accepted. To verify the assumptions of normality, linearity, homogeneity, and homoscedasticity, false regression analysis was selected as a multivariate technique that uses a set of theoretical random quantiles based on the χ^2 distribution to establish a comparison with the parameters and quantiles of the sample. The results of the false regression analysis are obtained through the histogram, Q–Q plot, and scatter plot presented in Figure 2.

Table 3. Descriptive statistics for the ordinal variables of the instrument applied to researchers by dimension.

Question	Min.	1st Quartile	Median	Mean	3rd Quartile	Max.	Standard Deviation
Factor 1: Economic Dimension							
p_4	1.000	2.000	3.000	3.188	4.000	5.000	1.1180
p_5	1.000	2.000	3.000	3.069	4.000	5.000	1.1093
p_6	1.000	3.000	3.000	3.342	4.000	5.000	1.1228
p_7	1.000	3.000	3.000	3.327	4.000	5.000	1.0530
p_8	1.000	3.000	3.000	3.427	4.000	5.000	1.0719
Factor 2: Social Dimension							
p_9	1.000	3.000	3.000	3.365	4.000	5.000	1.0945
p_{10}	1.000	3.000	3.000	3.215	4.000	5.000	1.0725
p_{11}	1.000	3.000	4.000	3.535	5.000	5.000	1.1931
p_{12}	1.000	3.000	3.000	3.196	4.000	5.000	1.0708
Factor 3: Environmental Dimension							
p_{13}	1.000	2.000	3.000	3.127	4.000	5.000	1.0954
p_{14}	1.000	2.750	3.000	3.165	4.000	5.000	1.1522
p_{15}	1.000	3.000	3.000	3.223	4.000	5.000	1.0491
p_{16}	1.000	2.000	3.000	3.196	4.000	5.000	1.1097
p_{17}	1.000	3.000	3.000	3.181	4.000	5.000	1.0892
p_{18}	1.000	3.000	4.000	3.758	5.000	5.000	1.1785
Factor 4: Institutional Dimension							
p_{19}	1.000	2.000	3.000	3.119	4.000	5.000	1.0677
p_{20}	1.000	2.000	3.000	3.015	4.000	5.000	1.1689
p_{21}	1.000	3.000	3.000	3.373	4.000	5.000	1.1403
p_{22}	1.00	3.00	3.00	3.15	4.00	5.00	1.0381
p_{23}	1.000	2.000	3.000	3.077	4.000	5.000	1.1227
p_{24}	1.000	3.000	3.000	3.196	4.000	5.000	1.0672
p_{25}	1.000	2.000	3.000	3.112	4.000	5.000	1.1041
p_{26}	1.000	2.000	3.000	3.100	4.000	5.000	1.1379

As can be seen in Figure 2a, when plotting the histogram of the standardized quantiles obtained through the regression model, it was observed that they were distributed uniformly, showing a distribution considerably similar to the normal distribution; therefore, the assumption of normality was accepted. In Figure 2b, the Q–Q plot is presented, where the sample quantiles were plotted against the theoretical quantiles, showing that the quantiles were arranged in a configuration quite close to a 45-degree straight line; thus, the assumption of linearity was accepted. Additionally, Figure 2c presents the scatter plot in which the standardized residuals of the sample were projected against the fitted residuals obtained through the false regression model. Through this procedure, it was observed that none of the quadrants presented an irregular proportion of quantiles relative to the others, and the distances of each quantile from the horizontal axis remained within similar ranges

in the interval from -2 to 2 . Moreover, no patterns or groupings were observed in the quantiles; therefore, the assumptions of homogeneity and homoscedasticity were accepted.

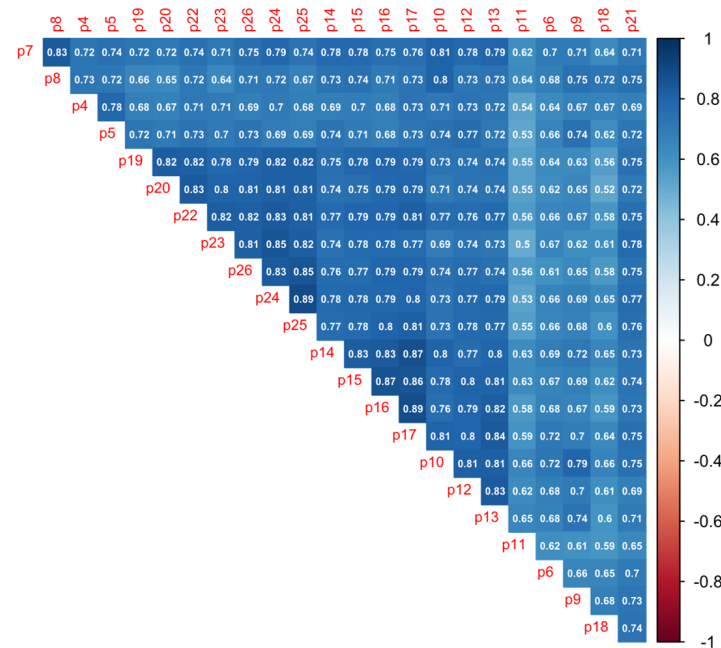


Figure 1. Multivariate correlation matrix for all pairs of variables of the instrument.

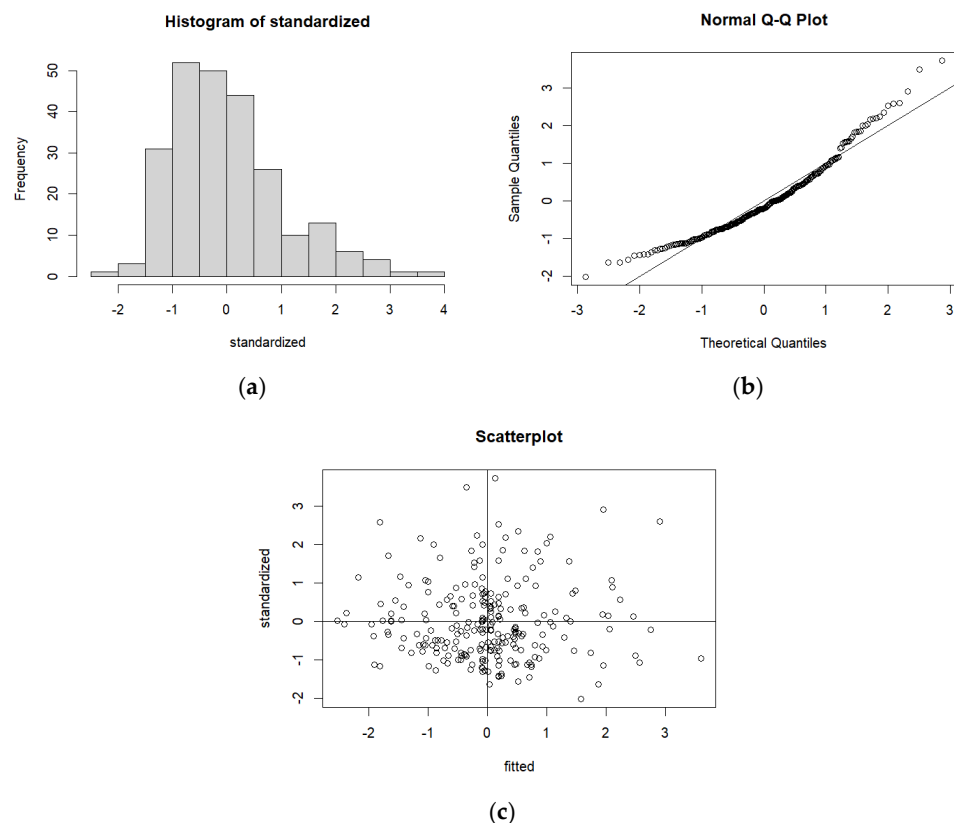


Figure 2. Results of the false regression analysis: (a) histogram, (b) Q-Q Plot, (c) ScatterPlot.

Once the assumptions were verified in the sample, CFA was applied by configuring its factorial structure using the lavaan library and the R statistical programming language (version 4.5.2). The results of the CFA are presented in Figure 3 and Table 4.

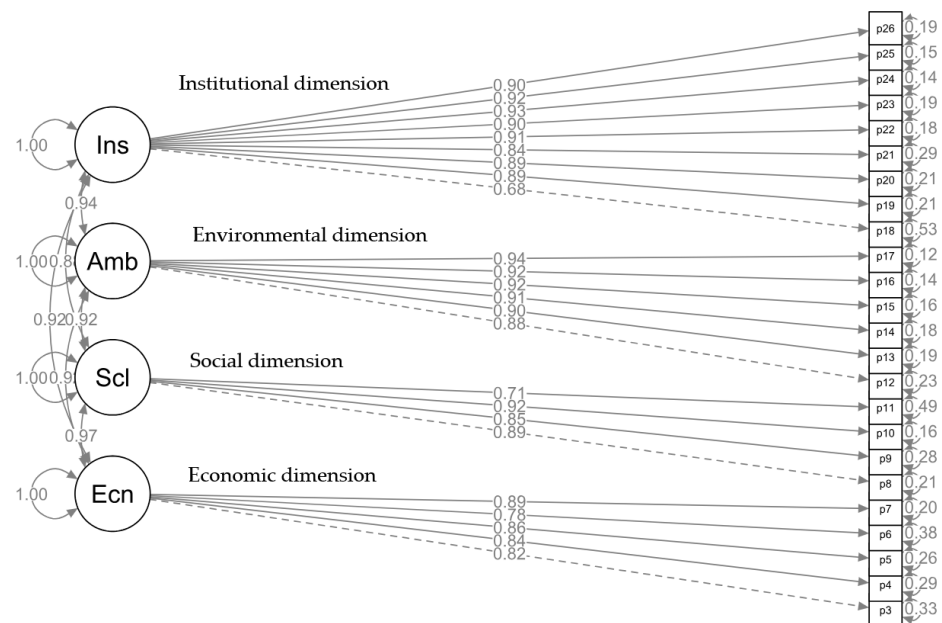


Figure 3. The figure illustrates the relative contribution of each observed variable to its corresponding sustainability dimension.

Table 4. Model fit indices for the structural equation model based on ordinal variables of the instrument applied to researchers.

Parameter	Value
Number of parameters (npar)	52
Minimum fit function (fmin)	1.294
Chi-square (χ^2)	626.196
Degrees of freedom (df)	224
<i>p</i> -value	0.000
Comparative Fit Index (CFI)	0.944
Tucker–Lewis Index (TLI)	0.937
Non-Normed Fit Index (NNFI)	0.937
Root Mean Square Error of Approximation (RMSEA)	0.046
Standardized Root Mean Square Residual (SRMR)	0.036

As shown in Figure 3, the standardized factor loadings of each item on its corresponding latent construct exceeded the threshold value of 0.50, indicating adequate convergent validity. Furthermore, the correlations among latent factors did not approach unity, suggesting the absence of multicollinearity and supporting discriminant validity within the factorial structure. Examination of the model goodness-of-fit indices revealed that the Comparative Fit Index (CFI), Tucker–Lewis Index (TLI), and Non-Normed Fit Index (NNFI) all exceeded 0.90, which, according to the literature [15], indicates an excellent model fit. Additionally, the Standardized Root Mean Square Residual (SRMR) and Root Mean Square Error of Approximation (RMSEA) values were 0.046 and 0.036, respectively, confirming the adequacy, validity, and reliability of the measurement instrument.

In a complementary manner, CFA allows the estimation of the coefficients of determination r^2 for each question within each factor. These coefficients can be interpreted as the level of influence that each question has in explaining its respective factor. In this way, the coefficients of determination made it possible to identify the components of the instrument that have

the greatest impact on each evaluated dimension of the research models with a sustainability focus. The coefficients of determination obtained through CFA are presented in Figure 4.

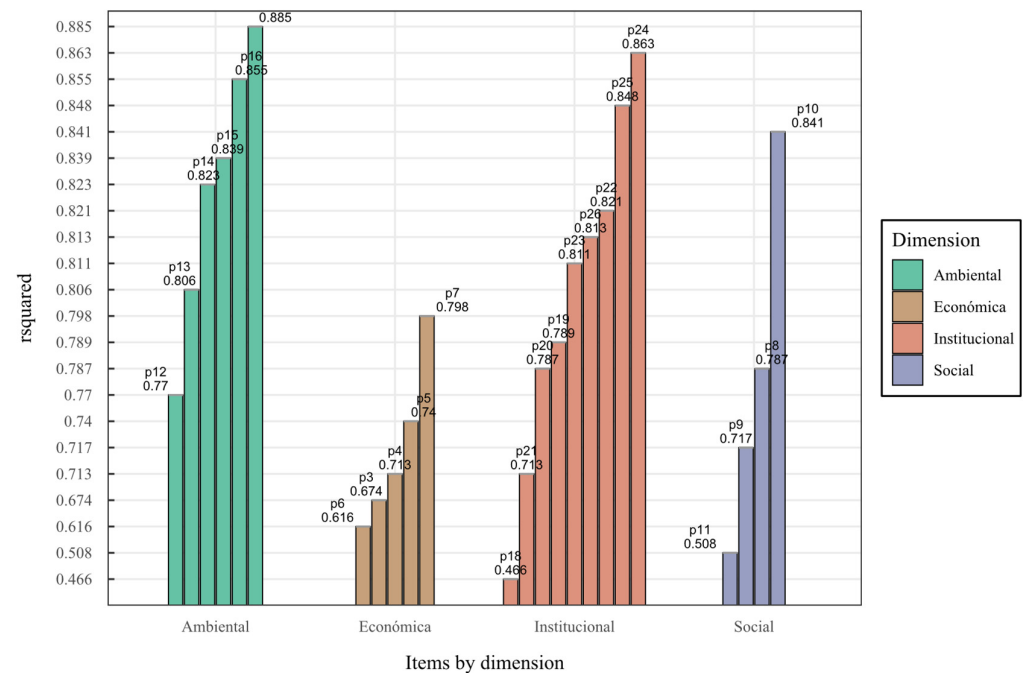


Figure 4. Coefficients of determination r^2 for each question within the dimensions of the factorial structure.

As can be observed in Figure 3, within the environmental dimension, question 17, “there is a significant percentage of research seedbeds that focus their work on the environmental dimension,” achieved the highest coefficient of determination, being the one that contributes the most to explaining the variance of that dimension’s eta. Similarly, question 7, “the HEI develops a considerable number of ad honorem sustainability-related research activities per year,” showed the greatest contribution to explaining the variance of the unobserved latent variable “economic dimension.” In the institutional dimension, it was observed that question 24, “there is an appropriate number of clubs focused on research and sustainability activities,” was perceived as the most important by the surveyed researchers, with a coefficient of determination of 0.863. Finally, in the social dimension, it was evidenced that question 10, “there is an adequate number of departments and/or centers that carry out research activities providing services to society,” had the highest determination index, being the most influential for this dimension.

Additionally, as mentioned in [35], these coefficients of determination can be used to create a regression model that allows the extraction of factor scores based on the contributions of each of the questions that compose them. In this way, Equations (12)–(15) were obtained, which allow the calculation of the observed score for each researcher in each dimension:

$$\xi_{económica} = 5.5202(0.692p_4 + 0.728p_5 + 0.625p_6 + 0.809p_7 + 0.769p_8), \quad (14)$$

$$\xi_{social} = 7.0746(0.693p_9 + 0.840p_{10} + 0.502p_{11} + 0.792p_{12}), \quad (15)$$

$$\xi_{ambiental} = 4.2607(0.799p_{13} + 0.830p_{14} + 0.837p_{15} + 0.854p_{16} + 0.890p_{17} + 0.484p_{18}) \quad (16)$$

$$\xi_{institucional} = 3.1003(0.791p_{19} + 0.793p_{20} + 0.705p_{21} + 0.824p_{22} + 0.809p_{23} + 0.863p_{24} + 0.850p_{25} + 0.816p_{26}), \quad (17)$$

where $\xi_{económica}$, ξ_{social} , $\xi_{ambiental}$ and $\xi_{institucional}$ are the scores for each evaluated dimension of the sustainability-oriented research models of the participating universities, observed for each of their researchers. In this way, the scores for each observation were weighted on a

scale from 0 to 100 and were compared with the scores obtained from the real indicators of each institution collected through the research departments of each university. The results are presented in Figure 5.

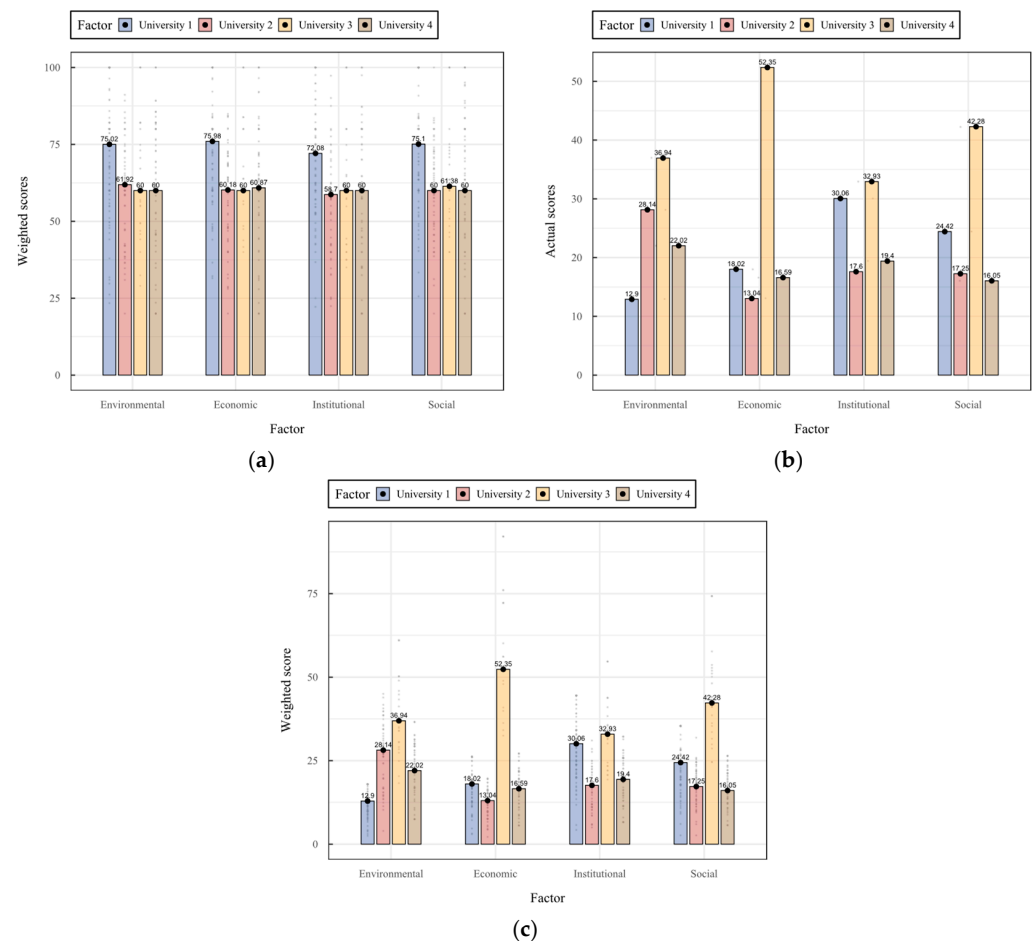


Figure 5. Comparative analysis of sustainability performance across the four participating universities. (a) Weighted sustainability scores by dimension derived from the confirmatory factor analysis (CFA). (b) Observed sustainability scores obtained from the survey responses. (c) Adjusted and normalized sustainability scores after centering and scaling.

As shown in Figure 5, the scores weighted through the confirmatory factor analysis (CFA) differ considerably from the scores reported by the research departments of each institution. This finding reveals a pronounced nonlinear dependence between researchers' perceptions and the actual institutional indicators. To obtain the final evaluation model for sustainability-oriented university research, a neural network-based approach was developed, enabling the model to account for data variance and the nonlinear relationships observed across institutions.

To this end, the procedure began by centering and scaling the data for each university and each sustainability dimension around the median value of each category represented in Figure 5b. This process yielded properly weighted and normalized scores for each dimension and each institution. The verification of the correct centering and scaling of the instrument results is presented in Figure 5c.

Subsequently, a sequential multilayer neural network model was designed to correctly obtain the score for each dimension based on each researcher's responses to the proposed survey. To this end, the dataset was first divided into a training dataset and a testing dataset, where questions p4 to p26 were configured as predictor variables and the scores of the unobserved latent variables for the environmental, economic, institutional, and social

dimensions were set as response variables. In this way, a neural network was configured with 26 neurons in its input layer corresponding to the 26 questions of the instrument and 4 neurons in the output layer corresponding to the four unobserved latent variables that reflect the score for each dimension obtained through the neural network.

Subsequently, Principal Component Analysis (PCA) was applied to determine the topology of the neural network, which, as established in [18], can be used to determine the optimal number of hidden layers that the neural network should have. The results of the PCA are presented in Figure 6 and Table 5.

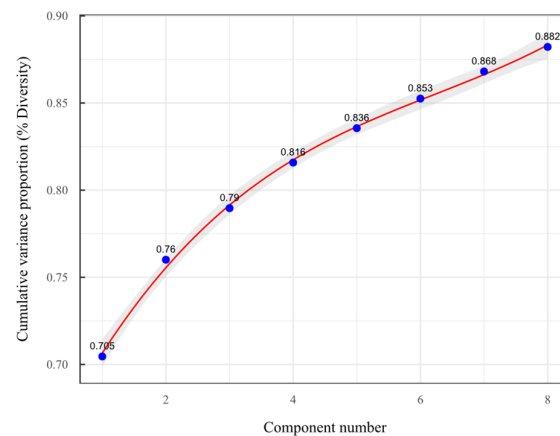


Figure 6. Principal component analysis for the instrument applied to researchers from the participating universities. The red line represents the cumulative explained variance trend across the principal components, while the blue circles indicate the observed values for each individual component.

Table 5. Results of the principal component analysis for the first five components, performed on the dataset.

Components	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5
Standard deviation	4.6190	1.2963	0.9457	0.8899	0.7730
Proportion of variance	0.7045	0.554	0.0295	0.0261	0.0197
Cumulative proportion	0.7045	0.7600	0.8158	0.8158	0.8355

As can be observed in Figure 6 and Table 5, the amount of variance explained by the first component is quite high, reaching a total of 70.45%, which can be explained by a single component. Based on the recommendations of [18], the optimal number of hidden layers that can be used for the design of a neural network is found at the component from which more than 80% of the cumulative variance is exceeded. In this way, the selected optimal number of hidden layers was three, since from the third principal component onward, more than 80% of the variance observed in the dataset is accumulated.

Subsequently, the number of neurons to be included in each hidden layer of the model was verified. For this purpose, the TensorFlow and Keras libraries were used, managed through the Anaconda environment and interfaced with R via the reticulate library. Following the recommendations of [23], it was considered appropriate to use from half to twice the number of neurons in the input layer for the first hidden layer, and similarly for the subsequent layers. In this way, to determine the topology of the first hidden layer of the neural network, 40 neural network models were trained and tested, assigning between 13 and 52 neurons to the first hidden layer, since the number of neurons in the input layer was 26, corresponding to each of the questions of the instrument applied to the 260 researchers from the studied universities. Similarly, 34 deep learning models were trained for the second hidden layer of the neural network, considering options ranging from 11 to 44 neurons; for the third hidden layer, 36 neural networks were trained with

configurations ranging from 11 to 46 neurons; and for the fourth hidden layer, 39 models were trained with configurations ranging from 12 to 50 neurons. The results obtained for each of the 149 trained models are presented in Table 6.

Table 6. Performance metrics for each of the designed neural network architectures.

First Hidden Layer											
Number of Neurons	Loss	Accuracy	Loss	Number of Neurons	Loss	Accuracy	mse	Number of Neurons	Loss	Accuracy	mse
13	6.61049	0.52000	143.97343	27	7.64279	0.52000	161.66096	41	7.47758	0.48000	154.70026
14	7.35568	0.56000	161.42709	28	7.74570	0.48000	180.97456	42	6.96407	0.52000	152.56285
15	6.78398	0.48000	152.78194	29	7.99225	0.52000	171.75630	43	7.81216	0.52000	175.81245
16	7.93419	0.52000	175.54967	30	7.54041	0.48000	164.20654	44	7.60679	0.56000	160.68207
17	7.46285	0.56000	163.58923	31	7.49249	0.48000	161.27461	45	7.50052	0.60000	160.53140
18	7.19733	0.44000	158.02576	32	7.43146	0.48000	157.93315	46	7.29040	0.52000	158.58780
19	7.59601	0.52000	155.03230	33	7.23824	0.56000	156.31580	47	7.77413	0.52000	162.27827
20	7.85943	0.48000	177.60406	34	7.45381	0.56000	161.67529	48	7.09082	0.52000	154.42505
21	7.38168	0.56000	153.87184	35	7.64535	0.56000	165.41623	49	7.78823	0.52000	159.25148
22	6.66107	0.60000	150.59218	36	7.38929	0.56000	166.78036	50	8.15013	0.52000	170.45750
23	7.33637	0.48000	163.91420	37	7.67119	0.36000	169.39720	51	7.48246	0.48000	157.51111
24	7.59702	0.52000	168.94106	38	7.46516	0.52000	156.48309	52	7.70194	0.52000	167.40285
25	7.39372	0.36000	156.18542	39	7.58592	0.56000	178.07089				
26	7.58095	0.40000	163.36514	40	7.32193	0.52000	161.51567				
Second hidden layer											
11	4.4676	0.4846	161.4548	23	4.9649	0.7154	156.4386	35	5.8384	0.5615	172.8735
12	5.6191	0.4846	188.0886	24	5.9431	0.5615	202.7327	36	6.0189	0.5615	179.3598
13	4.9793	0.7154	165.7875	25	6.0326	0.5615	199.2739	37	5.3896	0.4846	171.6785
14	4.6691	0.6385	169.0032	26	5.1963	0.4077	179.3626	38	5.5504	0.5615	186.1673
15	5.5874	0.4846	189.8610	27	5.2697	0.6385	183.9680	39	4.7814	0.4846	169.9322
16	5.0036	0.4077	171.2379	28	4.5833	0.4846	157.3014	40	5.8695	0.4846	174.7216
17	4.6083	0.6385	164.4200	29	6.3098	0.7154	177.2449	41	5.3071	0.6385	178.7738
18	5.4634	0.6385	165.2064	30	5.4610	0.4077	169.6887	42	6.1285	0.4077	207.8089
19	5.3136	0.6385	180.4027	31	5.3948	0.4077	175.9713	43	5.0295	0.6385	170.9867
20	6.0081	0.7154	190.3568	32	5.8722	0.5615	189.2480	44	5.1403	0.6385	177.5186
21	5.0952	0.6385	173.8194	33	5.8026	0.7154	179.2200				
22	5.4986	0.6385	172.5228	34	5.5039	0.6385	180.2224				
Third hidden layer											
11	2.3629	0.5246	94.5119	23	2.0884	0.5246	81.9454	35	2.7750	0.4477	94.2083
12	2.5910	0.6785	100.0968	24	3.1317	0.6015	103.1849	36	1.3735	0.7554	74.2720
13	2.8406	0.5246	104.8253	25	3.8011	0.9092	147.5736	37	2.5664	0.6015	104.5898
14	2.2003	0.7554	95.5288	26	1.4913	0.5246	74.6508	38	2.0267	0.5246	76.5736
15	1.9095	0.6015	73.6238	27	2.3569	0.6015	109.0628	39	3.6870	0.6015	132.1283
16	2.3928	0.6785	77.5403	28	2.6735	0.6785	95.8627	40	3.0592	0.4477	111.4355
17	2.0781	0.6015	60.6476	29	2.4002	0.6785	100.7877	41	2.0472	0.7554	73.2357
18	2.4656	0.6015	102.8611	30	2.4448	0.8323	110.6349	42	3.8664	0.8323	116.6717
19	3.1998	0.6015	116.5262	31	2.7592	0.6015	92.2450	43	2.1354	0.6015	103.2163
20	2.8175	0.6785	101.9881	32	2.7993	0.7554	100.6435	44	2.5400	0.3708	85.3244
21	2.3980	0.6785	81.7346	33	3.0743	0.5246	112.1566	45	2.3946	0.7554	92.1040
22	2.4792	0.5246	107.9653	34	2.4926	0.6785	95.8640	46	2.7339	0.6785	108.8427

Table 6. Cont.

First Hidden Layer											
Number of Neurons	Loss	Accuracy	Loss	Number of Neurons	Loss	Accuracy	mse	Number of Neurons	Loss	Accuracy	mse
Fourth hidden layer											
12	4.9775	0.6385	123.8721	25	7.1137	0.1769	165.8024	38	5.0995	0.5615	139.0243
13	5.3602	0.5615	121.5035	26	7.4176	0.1769	172.9749	39	5.5373	0.6385	145.3960
14	7.0724	0.1769	167.2437	27	5.7623	0.7154	144.1957	40	7.8832	0.1769	164.7424
15	5.6259	0.6385	139.3617	28	5.3984	0.4077	144.2599	41	5.7131	0.3308	149.5398
16	5.0447	0.5615	133.7712	29	5.2696	0.4077	127.5835	42	5.4261	0.5615	135.3566
17	5.0504	0.5615	125.1046	30	5.9735	0.1769	139.2516	43	6.0393	0.6385	170.7065
18	7.5365	0.1769	166.0810	31	21.8850	0.3308	667.6537	44	5.7384	0.3308	133.5269
19	4.8958	0.5615	139.3645	32	6.2996	0.6385	143.9705	45	8.6047	0.1769	154.6412
20	4.8681	0.6385	130.3944	33	7.6751	0.1769	163.2600	46	6.9127	0.1769	172.7332
21	4.5606	0.4077	117.0703	34	5.6771	0.5615	159.7961	47	5.4320	0.6385	151.8672
22	11.2039	0.1769	182.3223	35	5.5935	0.6385	125.8826	48	6.3469	0.5615	141.3358
23	5.8319	0.6385	154.5701	36	5.6255	0.5615	161.0143	49	5.7746	0.5615	147.5329
24	5.5590	0.5615	132.3280	37	5.4041	0.7154	151.3845	50	6.3801	0.5615	158.3410

Note: Bold values indicate the best-performing neural network configuration identified in this study.

As can be observed in Table 6, among the 149 neural network architectures tested, the architecture with 22, 23, and 25 neurons in its three hidden layers showed the best performance, achieving a loss of 3.8011, an accuracy of 0.9092, and an MSE of 147.5736. All trained models were designed using sequential models through the keras library, employing a normalization layer to scale the inputs into an appropriate format. SELU (Scaled Exponential Linear Unit) activation functions were used, along with an L_2 -type kernel regularizer with a coefficient $l = 0.001$. In addition, stochastic gradient descent was used as the optimization technique, with a learning rate of 0.1, a momentum of 0.8, and MAE (Mean Absolute Error) as the cost function. With this configuration and 156 training epochs for each model, it was possible to obtain a model capable of predicting, based on the survey, the score of each sustainability dimension with 90.92% accuracy, validated using the validation dataset that was never observed by the model during training.

Additionally, it can be noted that, although the principal component analysis suggested the use of three hidden layers, 39 additional neural networks with different configurations for a fourth hidden layer were implemented. However, it was observed that this fourth hidden layer did not improve accuracy compared to the three-hidden-layer model with the (22, 23, 25) configuration. Therefore, the findings from the PCA were confirmed, and the model with 26 input neurons, 22, 23, and 25 neurons in the three hidden layers, respectively, and four neurons in the output layer was selected as the best-performing configuration. The dataset, trained model, and supporting materials associated with this research are available in the Supplementary Materials section.

The training process of the best-performing models identified for each hidden layer is presented in Figure 7, while Figure 8 illustrates the architecture of the optimal artificial neural network configurations. The convergence of training metrics and the structural characteristics of the selected models confirm their capability to model complex nonlinear relationships in sustainability-oriented research evaluation.

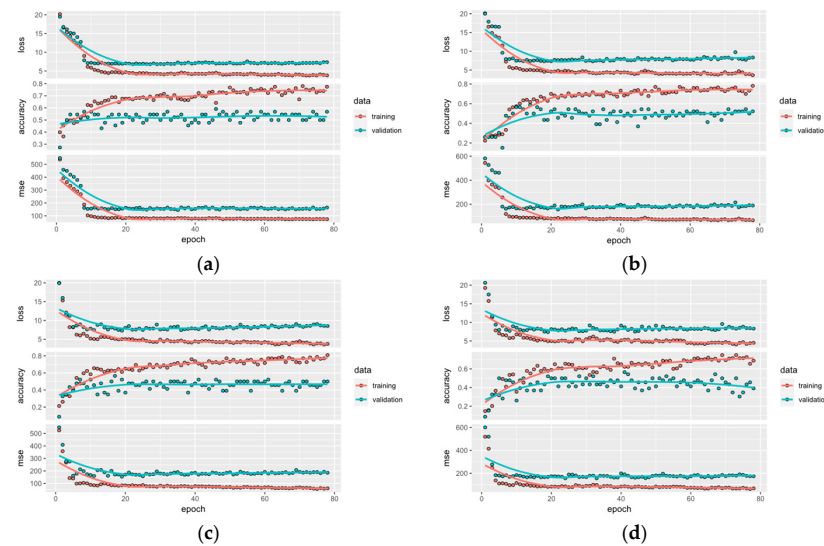


Figure 7. Training performance of the optimal artificial neural network (ANN) architectures across different hidden-layer configurations. (a) Model with one hidden layer composed of 22 neurons. (b) Model with two hidden layers composed of 22 and 23 neurons, respectively. (c) Model with three hidden layers composed of 22, 23, and 25 neurons, respectively. (d) Model with four hidden layers composed of 22, 23, 25, and 27 neurons, respectively.

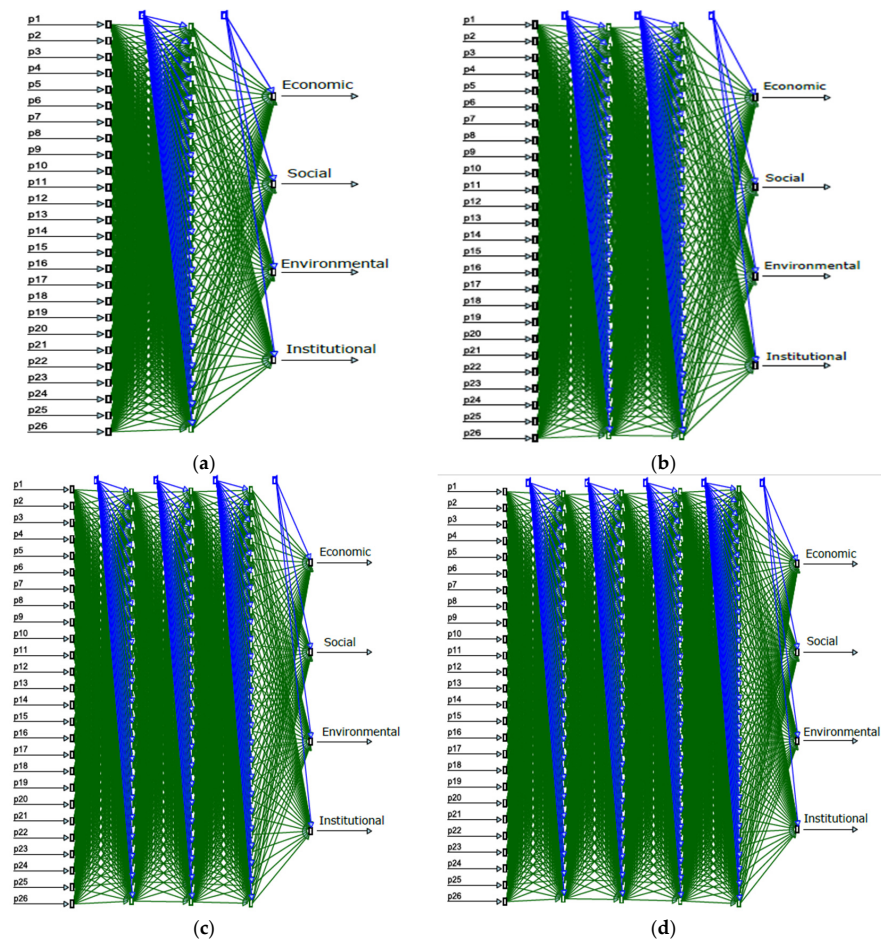


Figure 8. Architecture of the optimal artificial neural network (ANN) models across different hidden-layer configurations. (a) Model with one hidden layer composed of 22 neurons. (b) Model with two hidden layers composed of 22 and 23 neurons, respectively. (c) Model with three hidden layers composed of 22, 23, and 25 neurons, respectively. (d) Model with four hidden layers composed of 22, 23, 25, and 27 neurons, respectively.

4. Discussion

The present study differs significantly from previous related works in several aspects. First, unlike studies [2,24,25], which focused on highlighting the importance of university research for sustainable development, this study goes further by proposing and validating a concrete instrument to evaluate university research models in relation to sustainability indicators. In this way, it moves beyond the theoretical level and offers a practical tool that enables an objective diagnosis of the level of sustainability orientation of any higher education institution. Likewise, unlike the purely conceptual study of [9] on smart governance and the institutional analysis of [5] on ecological economics, this work incorporates solid empirical support by applying and validating the proposed instrument in four real universities in Ecuador and Colombia. Similarly, compared to literature reviews such as those in [7,16], the theoretical level is surpassed through a quantitative approach involving real participants.

In comparison with previous studies that proposed generic instruments such as those by [9,10] a distinguishing value of this study is that the developed instrument focuses specifically on university research models, exploring dimensions and indicators designed ad hoc for this purpose, rather than being an adaptation of broader multidimensional tools. This allows for a more detailed and relevant evaluation for the specific context of research systems in the HEIs.

In contrast to the instrument presented by [11] to evaluate environmental education in universities, this research incorporates a more comprehensive perspective by considering four dimensions: environmental, social, economic, and institutional. In this way, research models are evaluated holistically in relation to sustainability. Likewise, unlike the questionnaire focused on infrastructure aspects developed by [14], this study places the emphasis on research activities themselves and their outputs.

Another differentiating element is the robust statistical validation of the instrument through confirmatory factor analysis, regression, principal component analysis, and neural networks. This goes beyond the simple preliminary application of tools proposed in previous studies, providing the current instrument with greater reliability and offering a free, publicly accessible model that can be used to evaluate the research models of multiple universities in the region.

Regarding the results, a relevant finding was that researchers' perceptions did not correspond to the real indicators reported by the research departments. This motivated the development of a novel evaluation model based on neural networks, which achieved a 90.92% accuracy in predicting the real scores for each dimension. This deep learning model constitutes an important methodological contribution, as it allows the appropriate translation of subjective survey responses into objective scores aligned with the real metrics of each institution.

In this way, through methodological rigor, the proposal and validation of a specific instrument, the incorporation of a neural network model, and a quantitative approach with real participants, this study makes important theoretical and practical contributions. The findings represent an advance over previous research by providing an appropriate, free, and effective tool to diagnose and improve sustainability orientation in their respective research models.

5. Conclusions

This study proposed and validated a data-driven framework for assessing the sustainability orientation of research models in Higher Education Institutions, integrating environmental, social, economic, and institutional dimensions. The findings confirm that, although public HEIs show progress in incorporating sustainability principles into their

research systems, significant gaps persist between researchers' perceptions and actual institutional performance. These discrepancies highlight the limitations of traditional evaluation approaches and reinforce the need for more robust and analytically grounded sustainability assessment tools.

The multidimensional instrument developed in this research demonstrated strong validity and reliability, as evidenced by excellent goodness-of-fit indices obtained through Confirmatory Factor Analysis. This confirms that the indicators identified through expert judgment effectively capture the core dimensions required to evaluate sustainability-oriented university research models. Moreover, the analysis revealed that specific indicators—such as environmental research seedbeds, ad honorem research activities, research clubs, and units providing services to society—play a critical role in shaping sustainability performance and should be prioritized in institutional strategies.

One of the main methodological contributions of this study lies in the integration of artificial neural networks as a predictive component of the assessment framework. The proposed model achieved a predictive accuracy of 90.92%, demonstrating its capacity to model the non-linear relationships between subjective perceptions and objective institutional indicators. By translating survey data into reliable sustainability performance estimates, the framework overcomes a key limitation in sustainability assessment and provides a practical decision-support tool for university governance.

From a sustainability management perspective, the proposed framework enables Higher Education Institutions to identify performance gaps, prioritize interventions, and support continuous improvement processes in research governance. Its data-driven nature facilitates evidence-based decision-making and enhances institutional accountability in relation to sustainability objectives and the Sustainable Development Goals. Importantly, the framework is scalable and transferable, allowing adaptation to different institutional and territorial contexts beyond the Colombia–Ecuador border region.

Despite its contributions, this study has some limitations. The empirical application focused on a specific regional context and relied on cross-sectional data. Future research could extend the framework through longitudinal analyses, comparative studies across different higher education systems, and the integration of additional sustainability indicators. Furthermore, incorporating other artificial intelligence techniques may further enhance predictive performance and support more advanced sustainability governance models.

This research contributes to the advancement of sustainability assessment methodologies in higher education by offering a validated, data-driven, and practically applicable framework. By bridging measurement, analytics, and decision support, the study supports the development of more resilient, transparent, and sustainability-oriented university research systems.

Supplementary Materials: The dataset, trained model, and supporting materials used in this study are available at: <https://github.com/MarcoBurbanoPulles/InvestigationModelEvaluation> (18 February 2026).

Author Contributions: Conceptualization, M.R.B.P. and J.B.C.M.; methodology, J.B.C.M.; software, J.B.C.M.; validation, M.R.B.P. and J.B.C.M.; formal analysis, J.B.C.M.; investigation, M.R.B.P. and J.B.C.M.; resources, M.R.B.P.; data curation, J.B.C.M.; writing—original draft preparation, J.B.C.M.; writing—review and editing, M.R.B.P. and J.B.C.M.; visualization, J.B.C.M.; supervision, M.R.B.P.; project administration, M.R.B.P.; funding acquisition, J.B.C.M. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement: The study was approved by the Human Research Ethics Committee of Carchi State Polytechnic University (CEISH-UPEC) under protocol code CEISH-PI-001 on 11 September 2023. The research complied with all ethical, methodological, and legal requirements, and informed consent was obtained from all participants.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: <https://github.com/MarcoBurbanoPulles/statistical-data-processing/blob/main/enc-investigadores.csv>.

Conflicts of Interest: The authors declare no conflict of interest.

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